

Yuxin Xiao

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Research Interest

- **LLM Safety and Alignment:** Designing algorithms and investigating underlying mechanisms to ensure safe, robust, and interpretable LLM behaviors and trustworthy deployment in high-stakes domains.
- **LLM Reasoning and Decision-Making:** Advancing LLM reasoning, agentic planning, and deep research capabilities through reinforcement learning to address complex decision-making challenges in healthcare and beyond.

Education

Massachusetts Institute of Technology (MIT)

PH.D. IN SOCIAL & ENGINEERING SYSTEMS AND STATISTICS

Cambridge, MA

08/2022 - 12/2026

- Advised by **Prof. Marzyeh Ghassemi**; GPA: 5.00/5.00
- Affiliated with Institute for Data, Systems, and Society (IDSS) and Laboratory for Information and Decision Systems (LIDS)

Carnegie Mellon University (CMU)

M.S. IN MACHINE LEARNING

Pittsburgh, PA

08/2020 - 12/2021

- Advised by **Prof. Eric Xing** and **Prof. Louis-Philippe Morency**; GPA: 4.12/4.33

University of Illinois at Urbana-Champaign (UIUC)

B.S. IN COMPUTER SCIENCE; B.S. IN STATISTICS, MATHEMATICS

Urbana, IL

08/2016 - 05/2020

- Advised by **Prof. Jiawei Han** and **Prof. Hari Sundaram**; GPA: 3.93/4.00

Publications

(* indicates equal contribution)

Reinforcement Learning over Patient Trajectories for Clinical Reasoning in EHR Foundation Models

YUXIN XIAO, SHENG ZHANG, CHANDAN SINGH, TRISTAN NAUMANN, HOIFUNG POON, JIANFENG GAO, MARZYE H GHASSEMI, XIAODONG LIU

In Prep. 2025

[Paper], [Code]

KScope: A Framework for Characterizing the Knowledge Status of Language Models

YUXIN XIAO, SHAN CHEN, JACK GALLIFANT, DANIELLE BITTERMAN, THOMAS HARTVIGSEN, MARZYE H GHASSEMI

NeurIPS 2025

[Paper], [Code]

When Style Breaks Safety: Defending LLMs Against Superficial Style Alignment

YUXIN XIAO, SANA TONEKABONI, WALTER GERYCH, VINITH SURIYAKUMAR, MARZYE H GHASSEMI

Preprint 2025

[Paper], [Code]

MedBrowseComp: Benchmarking Medical Deep Research and Computer Use

SHAN CHEN, PEDRO MOREIRA, **YUXIN XIAO**, ET AL. (9 AUTHORS TOTAL)

Preprint 2025

[Paper], [Code]

Speak Easy: Eliciting Harmful Jailbreaks from LLMs with Simple Interactions

YIK SIU CHAN*, NARUTATSU RI*, **YUXIN XIAO***, MARZYE H GHASSEMI

ICML 2025

[Paper], [Code]

SFTMix: Elevating Language Model Instruction Tuning with Mixup Recipe

YUXIN XIAO, SHUJIAN ZHANG, WENXUAN ZHOU, MARZYE H GHASSEMI, SANQIANG ZHAO

Preprint 2024

[Paper], [Code]

In the Name of Fairness: Assessing the Bias in Clinical Record De-identification

YUXIN XIAO*, SHULAMMITE LIM*, TOM JOSEPH POLLARD, MARZYE H GHASSEMI

FAccT 2023

[Paper], [Code]

Uncertainty Quantification with Pre-trained Language Models: A Large-Scale Empirical Analysis

YUXIN XIAO, PAUL PU LIANG, UMANG BHATT, WILLIE NEISWANGER, RUSLAN SALAKHUTDINOV, LOUIS-PHILIPPE MORENCY

EMNLP 2022

[Paper], [Code]

SAIS: Supervising and Augmenting Intermediate Steps for Document-Level Relation Extraction

YUXIN XIAO, ZECHENG ZHANG, YUNING MAO, CARL YANG, JIAWEI HAN

NAACL 2022

[Paper], [Code]

Amortized Auto-Tuning: Cost-Efficient Bayesian Transfer Optimization for Hyperparameter Recommendation

YUXIN XIAO, ERIC P. XING, WILLIE NEISWANGER

Preprint 2021

[Paper], [Code]

Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark

CARL YANG*, **YUXIN XIAO***, YU ZHANG*, YIZHOU SUN, JIAWEI HAN

TKDE 2020

[Paper], [Code]

Discovering Strategic Behaviors for Collaborative Content-Production in Social Networks

YUXIN XIAO, ADIT KRISHNAN, HARI SUNDARAM

WWW 2020

[Paper], [Code]

Non-local Attention Learning on Large Heterogeneous Information Networks

YUXIN XIAO*, ZECHENG ZHANG*, CARL YANG, CHENGXIANG ZHAI

IEEE BigData 2019

[Paper], [Code]

Awards & Honors

2020 **C. W. Gear Outstanding Undergraduate Award**, UIUC (2 awardees at UIUC)

2020 **CRA Outstanding Undergraduate Researcher Award (Honorable Mention)**, CRA (135 awardees across North America)

Industry Research Experience

Microsoft Research

Bellevue, WA

RESEARCH INTERN @ DEEP LEARNING GROUP & HEALTH FUTURES

06/2025 - 08/2025

SUPERVISOR: **DR. XIAODONG LIU, DR. CHANDAN SINGH, DR. SHENG ZHANG, DR. TRISTAN NAUMANN**

Reinforcement Learning over Patient Trajectories for Clinical Reasoning in EHR Foundation Models

- Designed task-specific reward functions for clinical reasoning over longitudinal EHR data (MIMIC-IV), addressing challenges such as missing observations in patient trajectories and the need for accurate, time-sensitive medical predictions
- Developed and applied reinforcement learning algorithms to train EHR foundation models that reason over sequential clinical data, enabling a 9-million-parameter model to outperform GPT-5 on critical medical event prediction tasks.
- Advanced efficient and practical healthcare AI systems by demonstrating that compact, RL-trained models can achieve state-of-the-art clinical reasoning performance while remaining cost-effective and deployable for real-world clinical environments.

Zoom AI

San Jose, CA

RESEARCH INTERN @ GENAI R&D TEAM

06/2024 - 08/2024

SUPERVISOR: **DR. SHUJIAN ZHANG, DR. SANQIANG ZHAO**

SFTMix: Elevating Language Model Instruction Tuning with Mixup Recipe

- Designed and implemented SFTMix, a novel Mixup-based instruction-tuning algorithm that leverages training dynamics to identify confidence-based data subspaces and improve large language models' instruction tuning without curated datasets
- Conducted extensive experiments demonstrating that SFTMix consistently outperforms standard next-token prediction (NTP) across diverse instruction-following and healthcare-specific benchmarks
- Performed in-depth analysis of data confidence, regularization, and scalability, showing SFTMix's compatibility with data selection, parameter-efficient fine-tuning (LoRA), and compute-constrained deployment scenarios

Bosch Center for Artificial Intelligence

Stuttgart, Germany

RESEARCH INTERN @ NLP TEAM

06/2023 - 08/2023

SUPERVISOR: **PROF. ANNEMARIE FRIEDRICH, PROF. HEIKE ADEL**

InstructTransfer: Transfer Tuning of Human-Language Instructions for Black-Box Large Language Models

- Developed an automatic and cost-effective instruction-tuning strategy for black-box language models (e.g., ChatGPT) by transferring knowledge from prior tuning tasks via an optimal-transport-based task similarity kernel
- Outperformed existing tuning baselines on downstream natural language generation applications with only one-eighth of the tuning cost

Academic Research Experience

Healthy ML Group

MIT

SUPERVISOR: **PROF. MARZYEH GHASSEMI**

08/2022 - Present

KScope: A Framework for Characterizing the Knowledge Status of Language Models

- Proposed a five-status taxonomy of LLM knowledge based on the consistency and correctness of model responses, going beyond traditional conflict-based evaluations to characterize what a model knows about a given question more precisely
- Developed KScope, a hierarchical statistical testing framework that progressively refines hypotheses about LLM knowledge modes and assigns one of the five knowledge statuses
- Conducted large-scale evaluations across nine LLMs and four datasets, revealing key insights into how supporting context narrows knowledge gaps, how context features influence knowledge updates, and how targeted context summarization improves update effectiveness

When Style Breaks Safety: Defending Language Models Against Superficial Style Alignment

- Investigated how style patterns compromise LLM safety by defining and quantifying ASR inflation across 32 models and seven benchmarks
- Characterized the mechanisms behind increased vulnerability, demonstrating that ASR inflation correlates with models' relative attention to style patterns and their overlap with instruction-tuning data, and that superficial style alignment further amplifies jailbreak susceptibility
- Developed and validated SafeStyle, a defense strategy that augments a small amount of safety training data with matched style distributions, achieving consistent safety improvements across three models, six fine-tuning style settings, and two real-world instruction-tuning datasets

Speak Easy: Eliciting Harmful Jailbreaks from LLMs with Simple Interactions

- Investigated the real-world impact of jailbreak vulnerabilities by evaluating whether LLM responses are sufficiently actionable and informative to enable average users to carry out harmful actions
- Proposed HarmScore, a novel evaluation metric that quantifies how effectively jailbreak responses facilitate harmful behavior, offering a more practical perspective on LLM safety beyond traditional success rate metrics
- Developed Speak Easy, a multi-step, multilingual attack framework that exploits common interaction patterns, leading to substantial increases in jailbreak effectiveness and revealing critical, overlooked vulnerabilities in existing safety evaluations

In the Name of Fairness: Assessing the Bias in Clinical Record De-identification

- Examined the bias in de-identifying clinical discharge notes for patients of various demographic groups by conducting a large-scale empirical evaluation of nine popular de-identification baseline methods of different categories
- Prepared and annotated 100 note templates based on real-world clinical discharge notes and 16 name lists that were representative of different gender, race, name popularity, and the decade of popularity
- Investigated three factors that affected the de-identification quality and proposed an effective and method-agnostic solution by fine-tuning de-identification methods with clinical context and diverse names

MultiComp Lab

SUPERVISOR: **PROF. LOUIS-PHILIPPE MORENCY**

CMU

04/2021 - 04/2022

Uncertainty Quantification of Pre-trained Language Models: A Large-Scale Empirical Analysis

- Presented a holistic empirical analysis on how to compose a well-calibrated pre-trained language model-based prediction pipeline
- Examined how different pre-trained language models worked with various uncertainty quantifiers under distribution shifts and on diverse tasks
- Investigated the influence of pre-training datasets and strategies, model sizes, fine-tuning strategies, and uncertainty quantifiers on the calibration quality of pre-trained language models based on thousands of empirical observations
- Inspected the relationship between calibration quality and other aspects of model performance such as accuracy, sharpness, and robustness

SAILING Lab

SUPERVISOR: **PROF. ERIC XING**

CMU

09/2020 - 12/2021

Amortized Auto-Tuning: Cost-Efficient Bayesian Transfer Optimization for Hyperparameter Recommendation

- Proposed a multi-task multi-fidelity Bayesian optimization framework with a novel task kernel and acquisition function. Leveraged cheap-to-obtain low-fidelity observations to efficiently recognize promising hyperparameters for new tuning tasks via knowledge transfer
- Analyzed the cost-efficiency and flexibility of existing hyperparameter tuning baselines by surveying 36 methods based on seven specific criteria
- Computed a hyperparameter recommendation database to serve the research communities, which consisted of 27 unique image classification tuning tasks and 150 distinct configurations over a 16-dimensional nested hyperparameter space

Data Mining Group

SUPERVISOR: **PROF. JIAWEI HAN**

UIUC

03/2019 - 05/2021

SAIS: Supervising and Augmenting Intermediate Steps for Document-Level Relation Extraction

- Developed a multi-task learning framework for document-level relation extraction, which consisted of a broad spectrum of carefully designed tasks to explicitly teach pre-trained language models to capture the key sources of information—relevant contexts and entity types
- Boosted the model performance further via evidence-based data augmentation and ensemble inference, while preserving the computational cost by assessing the uncertainty of model predictions
- Achieved state-of-the-art relation extraction results on three benchmarks and outperformed the runner-up by 5.04% relatively in retrieving interpretable evidence for each extracted relation

Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark

- Analyzed and evaluated the performance of various types of heterogeneous network embedding models (proximity-preserving based, message-passing based, relation-learning based) on different applications (node classification, link prediction)
- Designed and implemented a unified and user-friendly experiment interface for fair and efficient comparison of 13 popular heterogeneous network embedding algorithms on four large-scale benchmark datasets; received over 260 stars and forks on the corresponding GitHub repository

Crowd Dynamics Lab

SUPERVISOR: **PROF. HARI SUNDARAM**

UIUC

10/2018 - 04/2020

Discovering Strategic Behaviors for Collaborative Content-Production in Social Networks

- Investigated the research question of whether resource-limited individuals were able to discover strategic behaviors associated with high payoffs when producing collaborative/interactive content in social networks
- Proposed a novel dynamic dual graph attention network which modeled individuals' content production strategies under social influence through a generative process
- Conducted a thorough qualitative analysis on a real-world social network with over seven million nodes and 400 million edges, which revealed three strong empirical findings about the emergence of individuals' strategies during the content production stage

Teaching Experience

Massachusetts Institute of Technology (MIT)

Cambridge, MA

TEACHING ASSISTANT & SESSION INSTRUCTOR

14.310x Data Science for Social Scientists, Instructors: *Prof. Esther Duflo, Prof. Sara Ellison*; Coordinator: *Dr. Karene Chu* 06/2022 - 08/2022

- Designed and lectured 13 weekly online recitations and office hours for 30 students from Aporta in Peru as a part of the MicroMasters Program in Statistics and Data Science (SDS)
- Launched a set of guidelines and marking rubrics to guide students' year-long data science project in the program for the local NGOs in Peru
- Developed a series of final exam questions on statistics, machine learning, and econometrics

University of Illinois at Urbana-Champaign (UIUC)

Urbana, IL

TEACHING ASSISTANT

CS446 Machine Learning, Instructor: *Prof. Sanmi Koyejo*

08/2019 - 12/2019

- Assisted in developing homework and exams, graded assignments for over 120 students
- Prepared and organized the final course project on the topic of automated machine learning

CS410 Text Information Systems, Instructor: *Prof. Chengxiang Zhai*

08/2019 - 12/2019

- Developed a software tool to intelligently analyze students' interactions and content contributions. Promoted students' online discussion on Piazza with the tool and evaluated their class participation more accurately and fairly

CS125 Intro to Computer Science, Instructor: *Prof. William Chapman*

01/2017 - 05/2017

- Guided students through lab activities during discussion sessions, solved students' problems during office hours